Controlled VLM alignment for grounded reasoning

same budget, same runner, same data caps - only the training signal changes

1 Problem

VLMs can answer correctly while using weak or wrong visual evidence

2 Controlled study

Same runner, same data caps, same evaluation path.

Qwen2-VL

QLoRA answer-only, rationale-generative, and explanation-aware runs

BLIP-2

Generative and contrastive-enhanced controls through Q-Former features

Answer-only

Rationale-generative

Explanation-aware

Contrastive-enhanced

3 Evaluation

Answer quality

Native dataset metric, not one universal score

Rationale quality

Measured only where gold rationales exist

Visual sensitivity

Masking checks whether answer scores move

4 Impact

Safer Interpretation

Accuracy, rationale quality and grounding are not merged.

Cleaner claims

Each claim is tied to a control, table, figure, or artifact.

Reusable protocol

The same study can be rerun by a student lab on one GPU.